ALICE L WALTON SCHOOL OF MEDICINE

Hematopoietic Stem Cell Transplantation

Last updated December 23, 2025 · 12 min read

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Learning Objectives (4)

After completing this brick, you will be able to:

1. Compare and contrast the three sources of hematopoietic stem cells.
2. Compare and contrast autologous and allogeneic hematopoietic stem cell transplantation.
3. Describe the transplant process, including important post-transplant m
4. Describe complications of hematopoietic stem cell transplantation.

CASE CONNECTION

A 35-year-old patient with a history of acute myeloid leukemia is undergoing evaluation for possible allogeneic stem cell transplant. They have questions about the transplant process. How would you walk them through this discussion?

Consider as you read, and we'll revisit this case at the end of this brick.

GO TO CONCLUSION ↓

What is Hematopoietic Stem Cell Transplantation?

Hematopoietic stem cell transplantation (HSCT), also known as stem cell transplant or bone marrow transplant, is the process of infusing hematopoietic stem cells (progenitor cells) to re-establish marrow function and, in some circumstances, provide an immunologic graft-versus-tumor effect. The use of HSCT is increasing, as it becomes safer and applicable to more diseases and as donor availability expands.

The first documented human marrow transplantation was performed in 1939 in a patient with gold-induced marrow aplasia; the patient was infused intravenously with marrow from a brother with an identical ABO blood type. Since then, as our knowledge of transplantation expands, HSCT has expanded to the point that 100, rmed worldwide in 2020.

Why use hematopoietic stem cells? Hematopoietic stem cells (HSCs) give rise to all elements of the blood (Figure 1). These are the cells that are required for successful engraftment in hematopoietic transplantations.

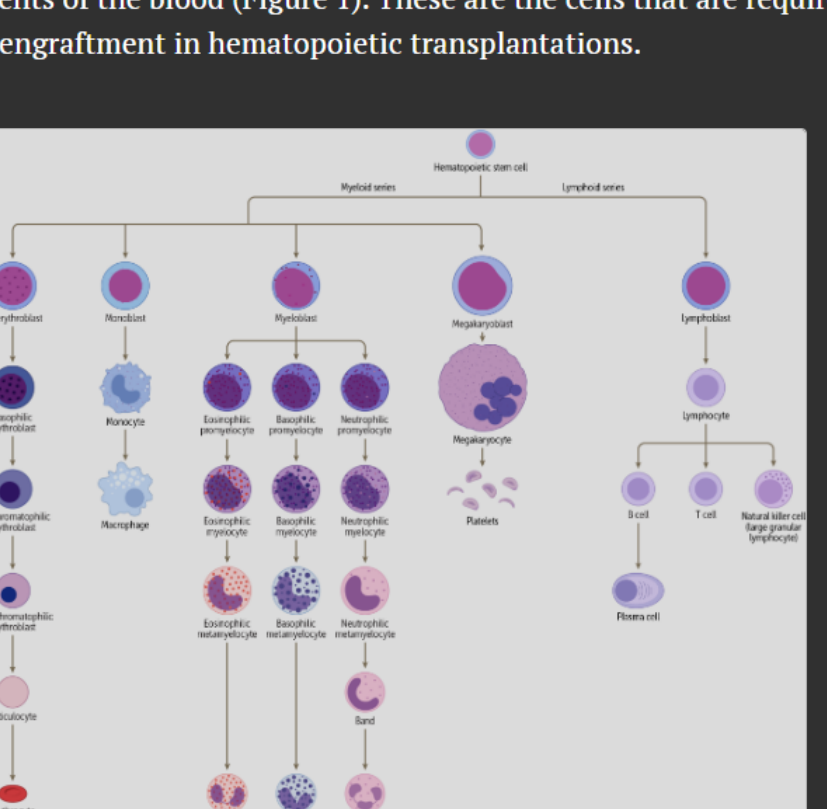


Figure 1

Sources of Hematopoietic Stem Cells (LO #1)

There are three main tly from the bone marrow, cells collected zation, and umbilical cord blood.

Bone Marrow

Bone marrow has historically been the traditional source for HSCs for transplant. The bone marrow procurement procedure is done under general anesthesia, in which marrow is aspirated through large-bore needles inserted into both sides of the posterior iliac crest (Figure 2). Marrow harvesting is generally a very safe procedure with minimal side effects.

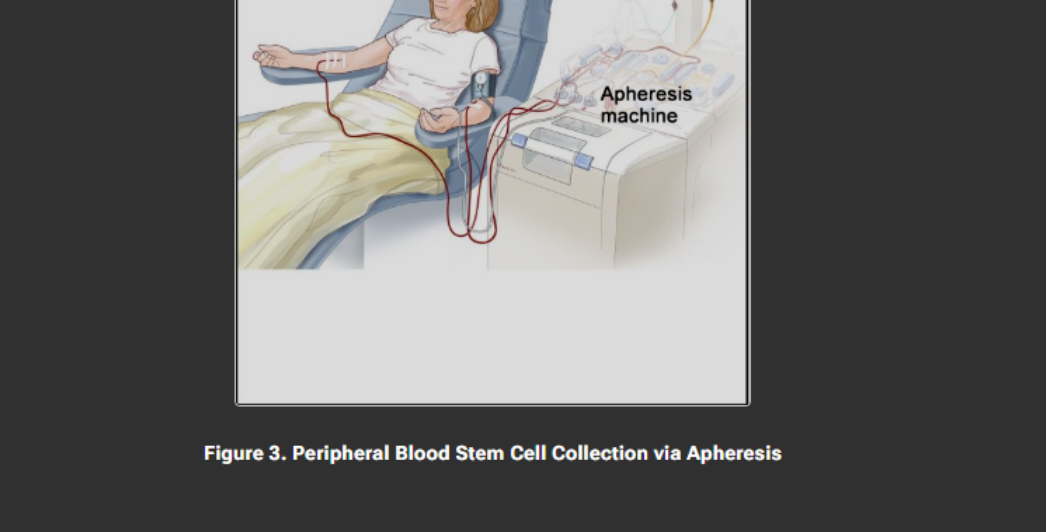


Figure 2. Bone Marrow Aspiration

Peripheral Blood

Normally, HSCs are present at very low levels in the blood. However, use of certain regimens can mobilize stem cells from the marrow into the blood. Once mobilized, HSCs can Typical agents used to mobilize HSCs include G-CSF (granulocyte colony stimulating factor) (most common), IL-3, thrombopoietin, and the CXCR4 antagonist plerixafor. This process is generally a very safe procedure, with the most common side effect being bone pain from the G-CSF administration.

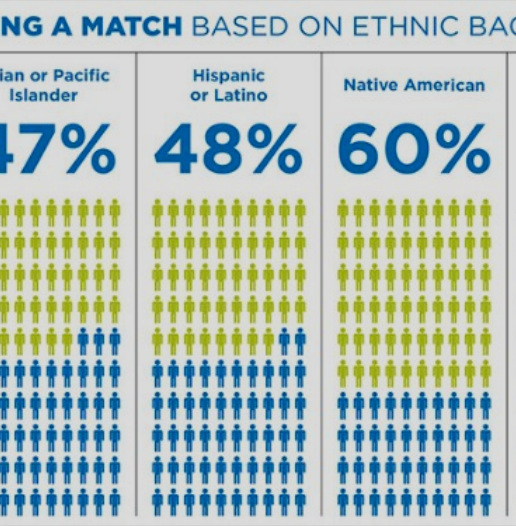

Blood-forming stem cells removed from donor
Stem cells
Apheresis machine

Figure 3. Peripheral Blood Stem Cell Collection via Apheresis

The adequacy of peripheral blood stem cell products is generally measured through the number of CD34+ cells per kilogram of recipient body weight. A threshold of greater than 2×10^6 CD34+ cells/kg is often considered the minimum acceptable roducts, although successful engraftment can occur with lower doses.

Umbilical Cord Blood

Umbilical cord blood (UCB) has emerged as an alternative-donor option for patients who lack HLA-identical siblings or suitably HLA-matched unrelated donors. This is particularly true for patients who belong to ethnic groups that are underrepresented in donor registries, those of mixed ethnicity, and those with uncommon HLA haplotypes (Figure 4).

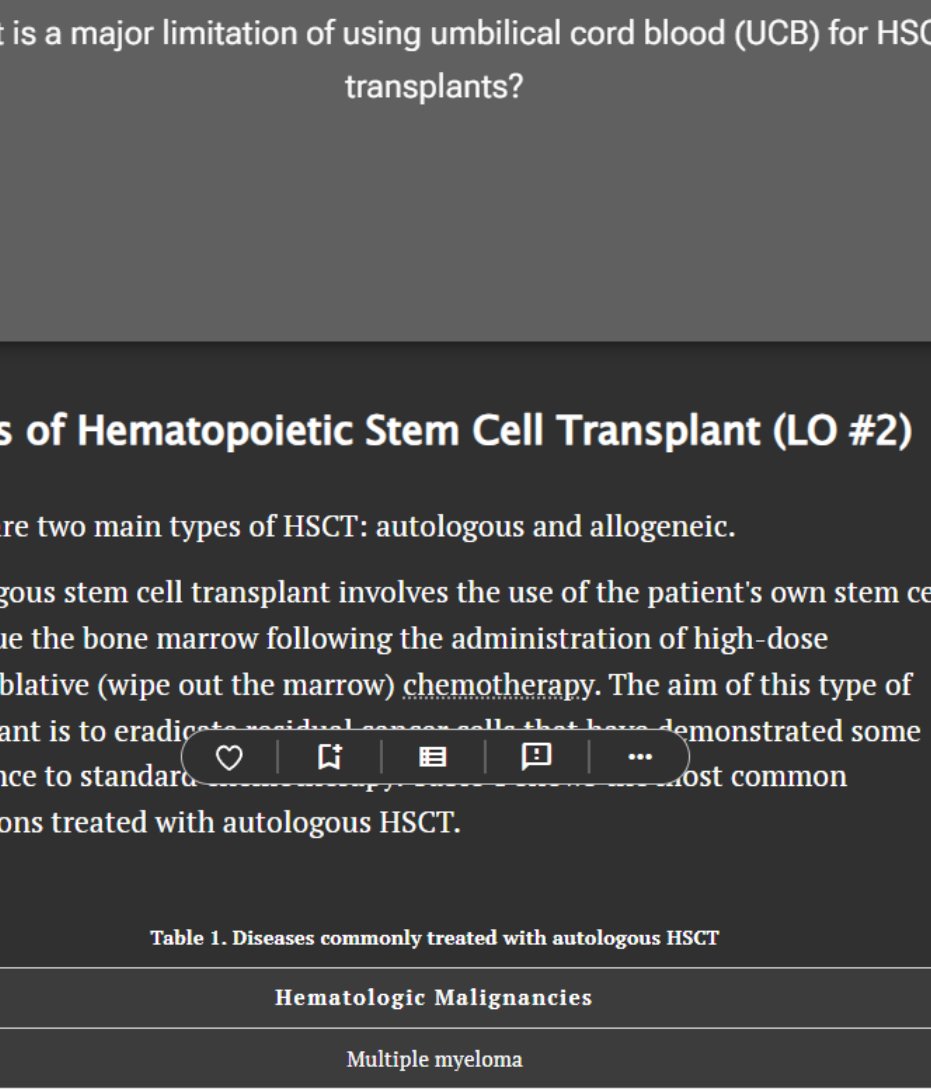


Figure 4

UCB is collected from the umbilical vessels in the placenta at the time of delivery and is a rich source of hematopoietic stem cells. Because these cells are immunologically histocompatibility barriers of convention

Major limitations of this method are the relatively small number of HSCs in each cord blood unit (cell dose requires 2 units/adult) and the increased risk of infections (particularly viral infections or reactivation), given the naive immunologic nature of the stem cells.

What is a major limitation of using umbilical cord blood (UCB) for HSC transplants?

Types of Hematopoietic Stem Cell Transplant (LO #2)

There are two main types of HSCT: autologous and allogeneic.

Autologous stem cell transplant involves the use of the patient's own stem cells to rescue the bone marrow following the administration of high-dose myeloablative (wipe out the marrow) chemotherapy. The aim of this type of transplant is to eradic demonstrated some resistance to standard ost common conditions treated with autologous HSCT.

Table 1. Diseases commonly treated with autologous HSCT

Hematologic Malignancies
Multiple myeloma
Non-Hodgkin lymphoma
Hodgkin lymphoma
Solid Tumors (e.g. neuroblastoma, germ cell tumors)
Autoimmune disorders (e.g. scleroderma)

Allogeneic stem cell transplant involves the infusion of stem cells from a related or unrelated donor resulting in two mechanisms for fighting the cancer. The mechanisms include the cytotoxic effects of the preparative chemotherapy as well as the donor's immune system attacking residual cancer cells (graft-versus-tumor effect).

Table 2. Diseases commonly treated with allogeneic HSCT

Hematologic Malignancies	Other Diseases
Acute myeloid leukemia	Aplastic anemia
Acute lymphoblastic leukemia	Paroxysmal nocturnal hemoglobinuria
Chronic myeloid leukemia	Fanconi Anemia
Myelodysplastic	emia major
Myeloproliferative neoplasms	Sickle cell anemia
Non-Hodgkin lymphoma	Severe combined immunodeficiency
Hodgkin lymphoma	Wiskott-Aldrich syndrome
Chronic lymphocytic leukemia (CLL)	Inborn errors of metabolism (i.e. Gaucher's disease)
Multiple myeloma	
Aplastic anemia	

Overview of the Transplant Process (LO #3)

Prior to the transplant, the recipient undergoes extensive evaluation, primarily to establish the patient's ABO and HLA type. Recall, ABO and HLA matching is critical to the success of transplantation, mitigating the risk of downstream rejection (refer to the prior brick on transplantation linked below for additional information).

Transplantation: https://exchange.scholarrx.com/brick/yog9mkee5qkp

HLA typing is accomplished through a blood draw or cheek swab. Once the HLA type is established, the transplant team endeavors to find a donor, first performing HLA typing on family members and searching the NMDP donor registry for potential unrelated donors or cord blood units.

CLINICAL CORRELATION

If you have a desire to become a bone marrow donor, the NMDP national registry (previously called B urces to join the registry.

https://www.nmdp.org/get-involved/join-the-registry/donate-bone-marrow

Regardless of the type of transplant, patients receiving HSCT undergo a similar process that is centered around the day of stem cell infusion, which is known as "Day 0". In the days prior to transplant, the patient receives conditioning chemotherapy, also known as the conditioning regimen (Figure 5). In general, the pretransplant conditioning regimens consist of a combination of chemotherapy drugs with or without radiation with the goal to immunosuppress the recipient and eliminate as much tumor as possible.

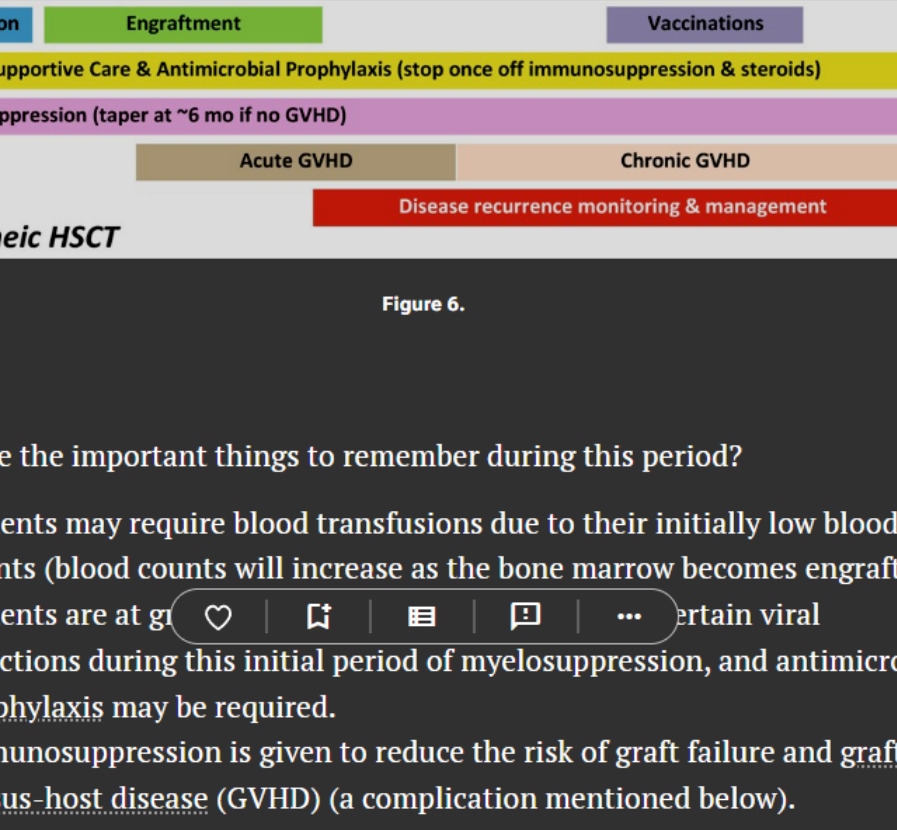

Autologous HSCT
CVC Placed
Conditioning Regimen
Stem Cell Infusion
Supportive Care & Antimicrobial Prophylaxis
PBSC Mobilization & Collection
Day -14
Day -10
Day -1
Day 0
Donor BM Harvest or PBSC Mobilization/Collection
arch
Supportive Care & Antimicrobial Prophylaxis
CVC Placed
Conditioning Regimen
Stem Cell Infusion
Immunosuppression
ic HSCT

Figure 5

The post-transplant period is known as the recovery period, associated with 2-4 weeks of myelosuppression from the conditioning regimen during which the donor cells are migrating to the bone marrow to begin producing new blood cells (Figure 6).

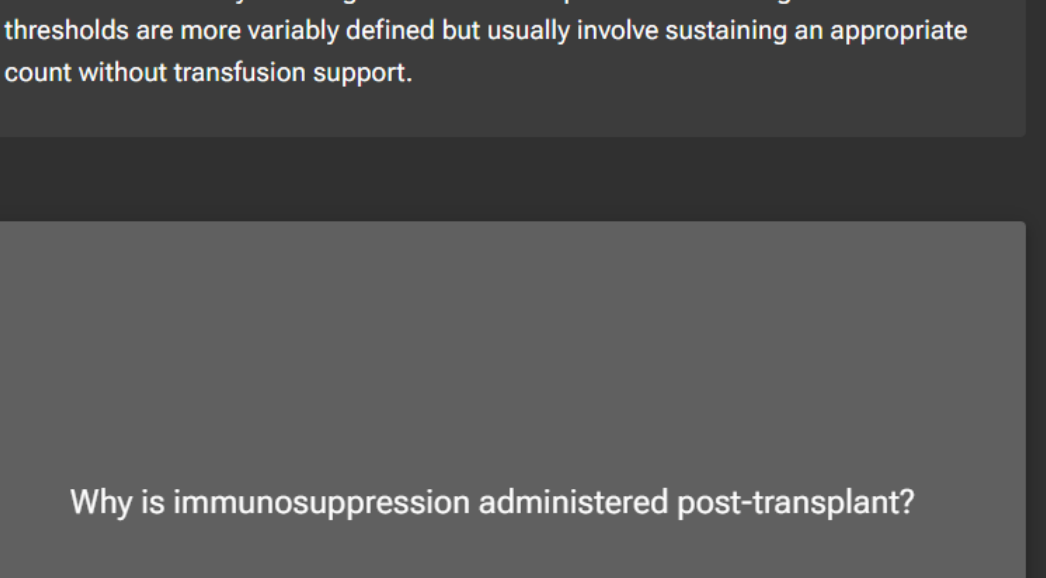

Autologous HSCT
Disease recurrence monitoring & management
Supportive Care & Antimicrobial Prophylaxis
Cell Infusion
Engraftment
Vaccinations
Day 0
Day +14
Day +30
Day +100
Day +180
1 year +
Cell Infusion
Engraftment
Vaccinations
Supportive Care & Antimicrobial Prophylaxis (stop once off immunosuppression & steroids)
Immunosuppression (taper at ~6 mo if no GVHD)
Acute GVHD
Chronic GVHD
Disease recurrence monitoring & management
Allogeneic HSCT

Figure 6.

What are the important things to remember during this period?

- Patients may require blood transfusions due to their initially low blood counts (blood counts will increase as the bone marrow becomes engrafted).
- Patients are at g ertain viral infections during this initial period of myelosuppression, and antimicrobial prophylaxis may be required.
- Immunosuppression is given to reduce the risk of graft failure and graft-versus-host disease (GVHD) (a complication mentioned below).

CLINICAL CORRELATION

Myeloid engraftment is defined as the first of 3 consecutive days in which the absolute neutrophil count exceeds 0.5×10^9/L. Myeloid engraftment typically occurs within 21 days of the graft infusion. The platelet and hemoglobin thresholds are more variably defined but usually involve sustaining an appropriate count without transfusion support.

Why is immunosuppression administered post-transplant?

Complications after Hematopoietic Stem Cell Transplantation (LO #4)

The first 100 days following transplantation are the time of greatest risk for transplant recipients. Some of the most commonly encountered complications will be discussed below.

Graft Failure

Graft failure is defined as the lack of donor hematopoietic cell engraftment (see criteria for engraftment above) following autologous or allogeneic HSCT. The overall incidence of graft failure following HSCT is low, 1-5% for autologous transplant and 5-6% for allogeneic transplant. Graft failure is divided into primary (early) and secondary (late) phases.

Primary graft failure is failure to achieve the threshold counts at any point beyond 28 days following HSCT. Secondary graft failure occurs in patients who initially meet threshold counts for engraftment but subsequently lose graft function in at least 2 cell lines.

Why would this occur? Graft failure has a myriad of causes, including graft rejection, persistent or progressive malignancy, low donor cell yield, medication side effects, medication, or GVHD.

CLINICAL CORRELATION

Graft rejection is a subset of primary or secondary graft failure and is caused by immune-mediated r nt immune cells.

What is the difference between primary and secondary graft failure?

Regimen-Related Organ Toxicities

Several well-known organ toxicities occur in hematopoietic stem cell transplant, usually due to the high-intensity conditioning regimen prior to transplant. A complete list is beyond the scope of this brick; however, one of the most common is oral mucositis.

Oral mucositis (inflammation of the mouth) occurs in a majority of transplant recipients (70-87% of patients in one study). Symptoms of oral mucositis include: a dry mouth, a sore mouth, and mouth ulcers (Figure 7).

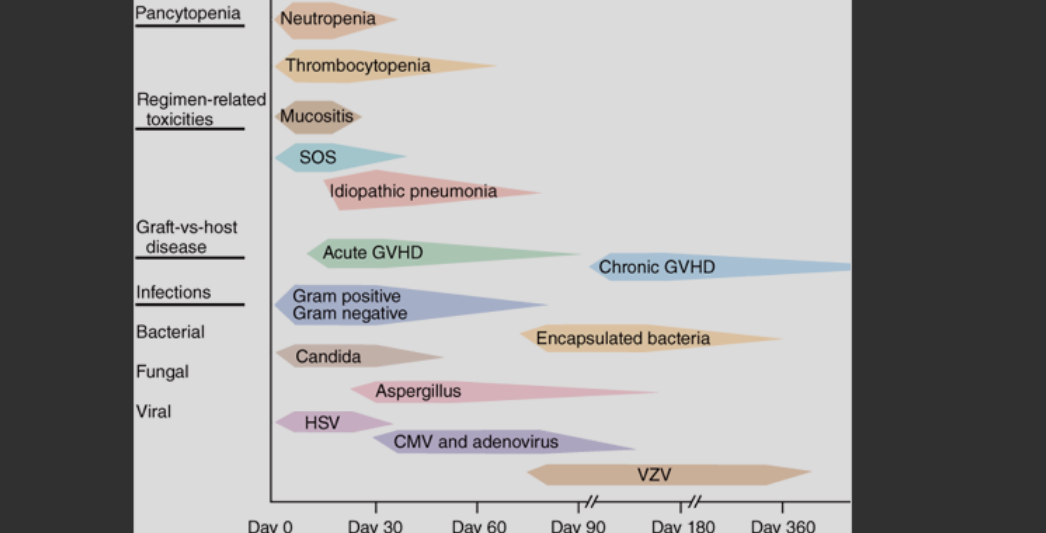


Figure 7. Oral mucositis. Right picture is accompanied by black hairy tongue.

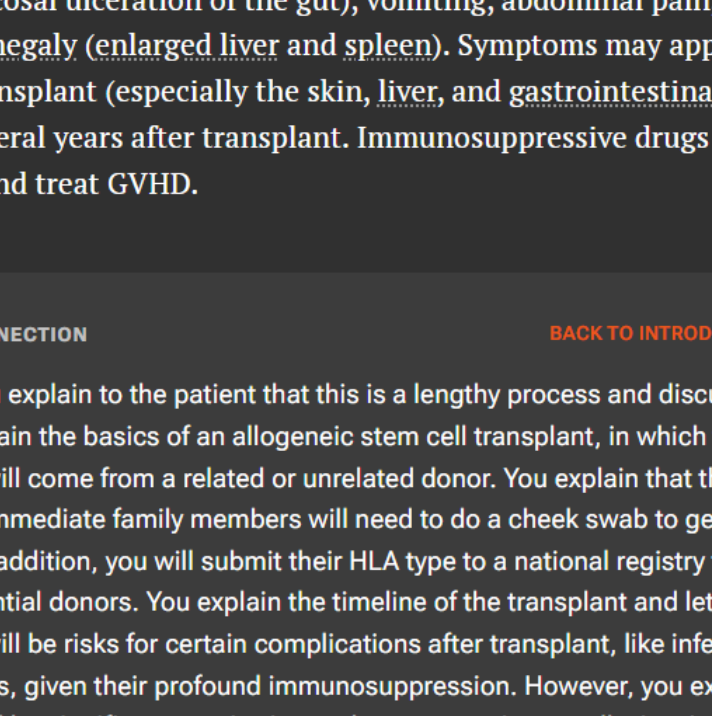
CLINICAL CORRELATION

Current management is supportive and includes rinsing with saline solutions and antimicrobials, cryotherapy, pain control, and possible parenteral (situated or occurring outside the mouth and intestine) nutrition.

Infection

Transplant recipients are at an increased risk for infection, given the profound immunosuppression.

Figure 8 shows common organisms associated with infection in the post-transplant period. For the purposes of this brick, just understand that the patient is at risk for bacterial, fungal, and viral infections.



Source: Joseph Loscalzo, Anthony Fauci, Dennis Kasper, Stephen Hauser, Dan Longo, J. Larry Jameson: Harrison's Principles of Internal Medicine, 21e Copyright © McGraw Hill. All rights reserved.

Figure 8. Major syndromes complicating marrow transplantation.

Graft-Versus-Host Disease

Graft-versus-host disease (GVHD) is a potentially fatal complication that typically occurs in transplantation of allogeneic bone marrow (but is rare in transplantation of other types of organs). In GVHD, immune cells from the *graft* recognize the *recipient's* tissue as foreign.

In GVHD, the donor's transplanted T cells attack the recipient's organs and tissues, resulting in severe organ dysfunction.

Patients with GVHD may present with a maculopapular rash (may lead to desquamation in seve kin and mucous membranes; caused by destruction of small bile ducts), nausea, diarrhea (caused by mucosal ulceration of the gut), vomiting, abdominal pain, and hepatosplenomegaly (enlarged liver and spleen). Symptoms may appear within weeks after transplant (especially the skin, liver, and gastrointestinal tract) or later, up to several years after transplant. Immunosuppressive drugs are used to both prevent and treat GVHD.

CASE CONNECTION BACK TO INTRODUCTION ↑

To begin, you explain to the patient that this is a lengthy process and discussion. You first explain the basics of an allogeneic stem cell transplant, in which the donor cells will come from a related or unrelated donor. You explain that they as well as any immediate family members will need to do a cheek swab to get their HLA type. In addition, you will submit their HLA type to a national registry to identify potential donors. You explain the timeline of the transplant and let them know there will be risks for certain complications after transplant, like infection and mucositis, given their profound immunosuppression. However, you explain that there will be significant monitoring and treatment given to alleviate/prevent these complications as much as possible.

Summary

- Hematopoietic stem cell transplantation (HSCT) is the process of infusing hematopoietic stem cells to re-establish marrow function and provide an immunologic graft-versus-tumor effect.
- Hematopoietic ste marrow, peripheral blood, and umbil
- There are two main types of HSCT: autologous, using the patient's own stem cells, and allogeneic, using stem cells from a related or unrelated donor.
- The transplant process involves extensive evaluation of the recipient, conditioning chemotherapy, stem cell infusion, and a recovery period.
- Myeloid engraftment, the process of the donor cells migrating to the bone marrow to begin producing new blood cells, typically occurs within 21 days of the graft infusion.
- Complications after HSCT can include graft failure, regimen-related organ toxicities, infection, and graft-versus-host disease (GVHD).

Review Questions

1. Which source of hematopoietic stem cells (HSCs) requires mobilization with G-CSF?

A. ○ Bone marrow
B. ● Peripheral Blood
C. ○ Umbilical cord blood

Hide Explanation

The correct answer is peripheral blood (B). G-CSF is used to mobilize stem cells from the bone marrow into the peripheral blood. Bone marrow (A) stem cells are acquired directly wit ood (C) is collected from the umbilical vessels in the placenta at the time of delivery.

2. A patient with a history of primary myelofibrosis is receiving a hematopoietic stem cell transplant from their younger sister. What type of transplant is being described?

A. ○ Autologous
B. ● Allogeneic

Hide Explanation

The correct answer is allogeneic stem cell transplant (B). Allogeneic stem cell transplant is one in which the patient receives stem cells from another person, either a related or unrelated donor. Stem cells can come from any source (bone marrow, peripheral blood, or umbilical cord blood). Autologous stem cell transplant (A) is one in which the patient receives their own stem cells (through bone marrow or peripheral blood collection).

3. What is given to reduce the risk of graft failure and GVHD?

A. ○ Antimicrobial prophylaxis
B. ○ Blood Transfusions
C. ○ Conditio
D. ● Immunosuppression

Hide Explanation

The correct answer is immunosuppression (D). Antimicrobial prophylaxis (A) is given to reduce the risk of infection. Blood transfusions (B) are used to treat profound anemia and/or thrombocytopenia. Conditioning regimen (C) is the treatment given before the transplant to wipe out the marrow to prepare them for the new stem cells.

4. What is a common organ toxicity that occurs in hematopoietic stem cell transplant due to the high-intensity conditioning regimen prior to transplant?

A. ○ Bone pain
B. ● Oral mucositis
C. ○ Jaundice
D. ○ Maculopapular rash

Hide Explanation

The correct answer is oral mucositis (B). Oral mucositis, or inflammation of the mouth, occurs in a m igh-intensity conditioning regimen mon side effect of G-CSF administration, which is used to mobilize stem cells from the marrow into the blood for peripheral blood stem cell collection. Jaundice (C) and maculopapular rash (D) are symptoms of graft-versus-host disease, a potential complication of allogeneic bone marrow transplantation.

How would you rate this Brick?

References

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- [redacted]
- [redacted]

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Compare and contrast the three sources of hematopoietic stem cells.~~
02 ☑ ~~Compare and contrast autologous and allogeneic hematopoietic stem cell transplantation.~~
03 ☑ ~~Describe the transplant process, including important post-~~

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Objective Confidence: 4/4